

ITEC617 – Modern Database Management

Assessment 3: Design Report

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Table of Contents

1.0 Introduction	1
2.0 Conceptual Design:.....	1
2.1 Entities and Relationships.....	1
2.1.1 Entities:	1
2.1.2 Relationships:	1
2.2 Determine appropriate attributes for each entity.	2
2.3 Entity-Relationship Diagram (ERD):.....	4
2.4 Normalization (3NF):	4
2.5 Identification of Datatypes and constraints	6
3. Write SQL Queries:.....	8
4. Database security and ethical issues	14
4.1 Ethical Considerations.....	14
4.2 Security Measures	15
4.3 Recommendations	16
Conclusion	16
Reference	17

1.0 Introduction

This paper explains the design of a database system that manages the most relevant information about volunteers, campaigns, donations, environmental resources, and climatic impacts an environmental NGO such as Greenpeace should consider. It aims at creating a centralized and, at the same time, efficient structure in such a way that the NGO could easily monitor the effectiveness of the different campaigns, evaluate volunteer participation, facilitate smooth operations, and take decisions based on the available resources. Important Entities Some properties are designed for volunteers, campaigns, donors, donations, environmental resources, climate impacts to maintain data integrity and support unambiguous linkages. The database supports the NGO purpose of better organizing data from significant projects on the environment and climate.

2.0 Conceptual Design:

2.1 Entities and Relationships

2.1.1 Entities:

- Volunteer
- Campaign
- Donor
- Donation
- Environmental Resource
- Climate Impact.

2.1.2 Relationships:

- **Many-to-Many:**
 - Volunteer - Campaign
 - Donor - Campaign
 - Campaign - Environmental Resource
 - Environmental Resource - Climate Impact

- **One-to-Many:**
 - Donor – Donation
 - Campaign – Donation

2.2 Determine appropriate attributes for each entity.

The basis of any robust database design must involve the judicious selection of attributes for each object to enable the business process, capture qualities that are important, establish uniqueness, and provide relationships between entities (Pinto et al., 2022).

The key steps include:

- Every entity should capture data that is needed to control fundamental business activities. To control scheduling and communications, for instance, a volunteer entity could have the following attributes for VolunteerID, Name, and Email.
- Characteristics of an entity must be representative of the inherent characteristics of an entity. A Campaign entity may possess properties like CampaignID, Name, Description, StartDate, and EndDate, in case a campaign must be tracked for its details.
- Each entity should have an identifier, often referred to as a primary key, to avoid duplication and to ensure the fetching of data is appropriate. In the Donor entity, DonorID provides an identification that uniquely identifies donors.
- Foreign keys are features that link connected items. For instance, the DonorID and CampaignID attributes by linking the gifts with specific donors and specific campaigns allow for effective reporting and increase the flow of data within a Donation entity.

Here are attributes for each entity in the database:

1. Volunteer

- VolunteerID (Primary Key)
- Name
- Email

2. Campaign

- CampaignID (Primary Key)

- Name
- Description
- StartDate
- EndDate
- FundRaisingGoal

3. Donor

- DonorID (Primary Key)
- Name
- Email

4. Donation

- DonationID (Primary Key)
- DonorID (Foreign Key)
- CampaignID (Foreign Key)
- Amount
- Date

5. EnvironmentalResource

- ResourceID (Primary Key)
- ResourceType
- Description

6. ClimateImpact

- ImpactID (Primary Key)
- ImpactType
- Description

2.3 Entity-Relationship Diagram (ERD):

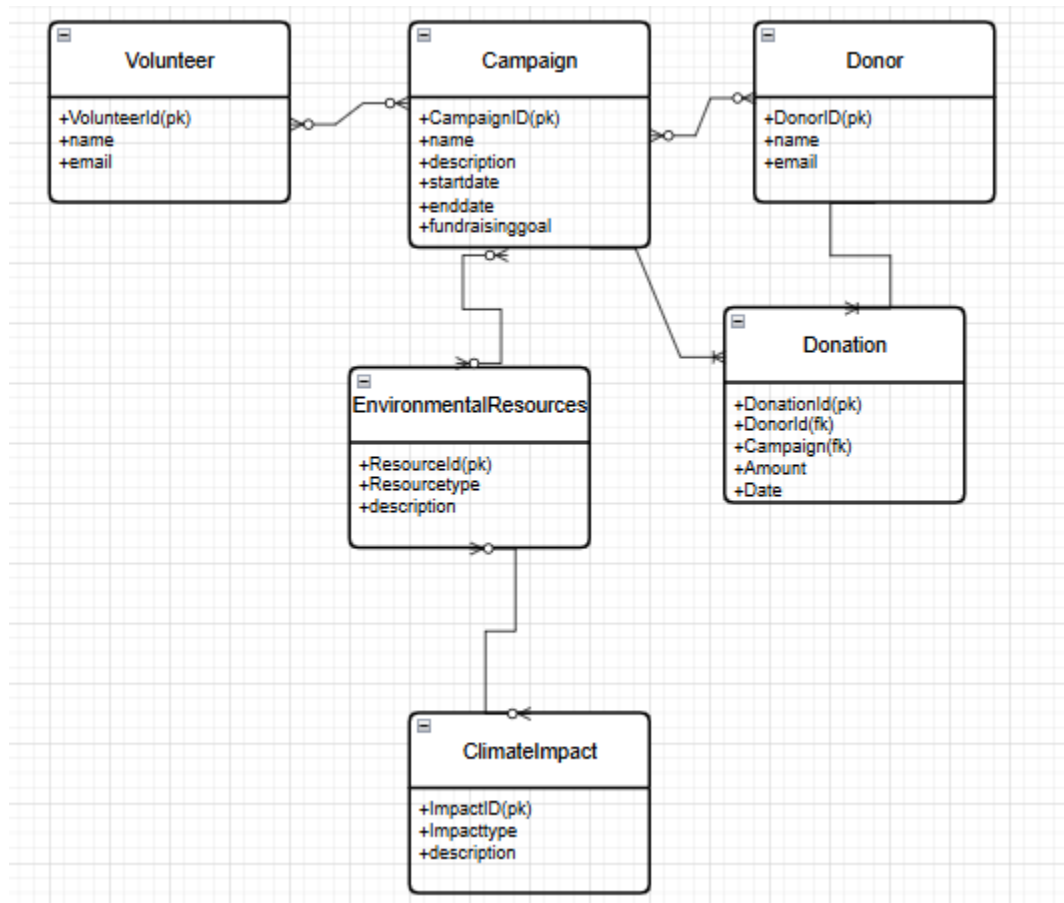


Fig.3. ERD for the database.

2.4 Normalization (3NF):

1. Volunteer

Volunteer
VolunteerID (Primary Key)
Name
Email

Table 4.1. 3NF table for Volunteer.

2. Campaign

Campaign
CampaignID (Primary Key)
Name
Description
StartDate
EndDate
FundRaisingGoal

Table 4.2. 3NF table for Campaign.

3. Donor

Donor
DonorID (Primary Key)
Name
Email

Table 4.3. 3NF table for Campaign.

4. Donation

Donation
DonationID (Primary Key)
DonorID (Foreign Key)
CampaignID (Foreign Key)
Amount
Date

Table 4.4. 3NF table for Donation.

5. EnvironmentalResource

Environmental Resource
ResourceID (Primary Key)
ResourceType
Description

Table 4.5. 3NF table for Campaign.

6. ClimateImpact

Climate Impact
ImpactID (Primary Key)
ImpactType
Description

Table 4.2. 3NF table for Campaign.

- Every non-prime attribute
- any attribute which is not a member of any primary key
- depends directly on the primary key. there are no transitive dependencies.
- No partial dependency A non-prime attribute is completely and fully dependent on the primary key.
- This prevents anomalies such as update and deleting anomalies and ensures integrity of data.

2.5 Identification of Datatypes and constraints

1. Volunteer

Attribute	Data Type	Constraint
VolunteerID	INT	Primary Key
Name	VARCHAR	100
Email	VARCHAR	255

Table 5.1. Identification of Datatypes and constraints for Volunteer entity.

2. Campaign Entity

Attribute	Data Type	Constraint
CampaignID	INT(10)	Primary Key
Name	VARCHAR2(100)	Not null
Description	TEXT(500)	Not null
StartDate	DATE	Not null
EndDate	DATE	Not null
FundRaisingGoal	DOUBLE (12,2)	Not null

Table 5.2. Identification of Datatypes and constraints for Campaign entity.

3. Donor Entity

Attribute	Data Type	Constraint
DonorID	INT(10)	Primary Key
Name	VARCHAR2(100)	Not null
Email	VARCHAR2(255)	Not null

Table 5.3. Identification of Datatypes and constraints for Donor entity.

4. Donation Entity

Attribute	Data Type	Constraint
DonationID	INT(10)	Primary Key
DonorID	INT(10)	Foreign Key
CampaignID	INT(10)	Foreign Key
Amount	DECIMAL(10,2)	Not null
Date	DATE	Not null

Table 5.4. Identification of Datatypes and constraints for Donation entity.

5. EnvironmentalResource Entity

Attribute	Data Type	Constraint
ResourceID	INT(10)	Primary Key
ResourceType	VARCHAR2(50)	Not null
Description	TEXT	Not null

Table 5.5. Identification of Datatypes and constraints for EnvironmentalResource entity.

6. ClimateImpact Entity

Attribute	Data Type	Constraint
ImpactID	INT(10)	Primary Key
ImpactType	VARCHAR(50)	Not null
Description	TEXT	Not null

Table 5.6. Identification of Datatypes and constraints for ClimateImpact entity.

3. Write SQL Queries:

1. Top Donors in a Campaign:

```
Query 1 x
SELECT d.DonorID, do.Name, SUM(d.Amount) AS TotalDonation
FROM Donation d
JOIN Campaign c ON d.CampaignID = c.CampaignID
JOIN Donor do ON d.DonorID = do.DonorID
WHERE c.Name = 'Ocean Conservation Campaign'
GROUP BY d.DonorID
ORDER BY TotalDonation DESC;
```

Output:

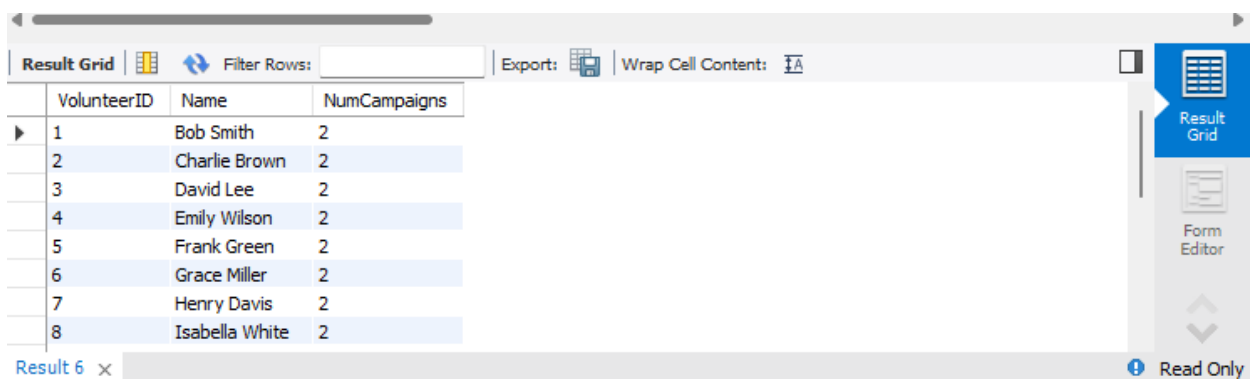
	DonorID	Name	TotalDonation
▶	15	Jacob Davis	420.00
	11	William Harris	300.00
	17	Benjamin Garcia	260.00
	3	Michael Johnson	240.00
	9	Benjamin Martin	220.00
	20	Charlotte Lee	200.00
	12	Charlotte Lewis	180.00
	18	Olivia Nelson	160.00
	8	Olivia Taylor	120.00
	14	Ava Jones	110.00
	5	David Lee	100.00

(Oracle, 2024)

2. Take out the number of campaigns each volunteer has participated in

```
1 • SELECT v.VolunteerID, v.Name, COUNT(d.DonationID) AS NumCampaigns
2 FROM Volunteer v
3 JOIN Donation d ON v.VolunteerID = d.DonorID
4 GROUP BY v.VolunteerID, v.Name;
```

Output:



The screenshot shows a database interface with a query result grid. The grid has three columns: VolunteerID, Name, and NumCampaigns. It contains 8 rows of data. The interface includes a toolbar with icons for various actions, a 'Filter Rows' field, an 'Export' button, and a 'Wrap Cell Content' option. On the right side, there are buttons for 'Result Grid' and 'Form Editor'. At the bottom, it says 'Result 6' and 'Read Only'.

VolunteerID	Name	NumCampaigns
1	Bob Smith	2
2	Charlie Brown	2
3	David Lee	2
4	Emily Wilson	2
5	Frank Green	2
6	Grace Miller	2
7	Henry Davis	2
8	Isabella White	2

3. The campaigns that are still accepting donations

```
1 • SELECT c.CampaignID, c.Name
2 FROM Campaign c
3 WHERE c.EndDate > CURDATE();
4
```

Output:

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
CampaignID	Name			
NULL	NULL			

4. Show the donors who have donated to multiple campaigns

```

1 • SELECT d.DonorID, COUNT(DISTINCT d.CampaignID) AS NumCampaigns
2   FROM Donation d
3  JOIN Donor do ON d.DonorID = do.DonorID
4  GROUP BY d.DonorID
5  HAVING COUNT(DISTINCT d.CampaignID) > 1
6  LIMIT 0, 1000;

```

Output:

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
DonorID	NumCampaigns		
1	2		
2	2		
3	3		
4	2		
5	2		
6	3		
7	2		
8	2		
9	2		
10	2		
14	2		

5. Calculate the average donation amount per donor across all campaigns they have donated to

```

1 • SELECT d.DonorID, AVG(d.Amount) AS AverageDonation
2 FROM Donation d
3 GROUP BY d.DonorID;

```

Output:

Result Grid		
Filter Rows:		
Export:		
Wrap Cell Content:		
	DonorID	AverageDonation
▶	1	110.000000
	2	55.000000
	3	226.666667
	4	168.333333
	5	110.000000
	6	128.333333
	7	253.333333
	8	190.000000
	9	183.333333
	10	100.000000
	11	360.000000
	12	180.000000
	13	270.000000
	14	122.500000
	15	385.000000
	17	290.000000
	18	145.000000
	19	440.000000
	20	200.000000

6. Retrieve the number of campaigns that have been affected by each climate impact

```

1 • SELECT ci.ImpactType, COUNT(c.CampaignID) AS AffectedCampaigns
2 FROM ClimateImpact ci
3 JOIN Campaign c ON EXISTS (
4     SELECT 1
5     FROM Donation d
6     WHERE d.CampaignID = c.CampaignID
7     AND d.Amount > 0
8     AND (c.Description LIKE '%climate%' OR c.Description LIKE '%environmental%')
9 )
10 AND c.Description LIKE CONCAT('%', ci.ImpactType, '%')
11 GROUP BY ci.ImpactType;

```

Output:

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
ImpactType	AffectedCampaigns		
Climate Change	1		

7. The volunteers who have participated in campaigns and raised more than 100.

```

1 • SELECT v.VolunteerID, v.Name
2 FROM Volunteer v
3 JOIN Donation d ON v.VolunteerID = d.DonorID
4 JOIN Campaign c ON d.CampaignID = c.CampaignID
5 WHERE (
6     SELECT SUM(d.Amount)
7     FROM Donation d2
8     WHERE d2.CampaignID = c.CampaignID
9 ) > 100
10 GROUP BY v.VolunteerID, v.Name;

```

Output:

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
VolunteerID	Name		
1	Bob Smith		
4	Emily Wilson		
7	Henry Davis		
10	Kevin Nelson		
6	Grace Miller		
19	Tanya Carter		
2	Charlie Brown		
5	Frank Green		
8	Isabella White		
11	Laura Mitchell		
14	Oliver Wright		
17	Rachel Moore		
3	David Lee		

8. The campaigns that have not yet met their fundraising goals and have less than a certain number of days remaining until the end date

```

1 • SELECT c.CampaignID, c.Name, c.FundraisingGoal,
2       (SELECT SUM(d.Amount) FROM Donation d WHERE d.CampaignID = c.CampaignID) AS TotalRaised,
3       DATEDIFF(c.EndDate, CURDATE()) AS DaysRemaining
4 FROM Campaign c
5 WHERE (SELECT SUM(d.Amount) FROM Donation d WHERE d.CampaignID = c.CampaignID) < c.FundraisingGoal
6 AND DATEDIFF(c.EndDate, CURDATE()) < 2;

```

Output:

Result Grid Filter Rows: Export: Wrap Cell Content:				
CampaignID	Name	FundraisingGoal	TotalRaised	DaysRemaining

9. The top 10 resources that have the greatest impact on climate change, based on the number of campaigns they have been associated with

```

1 • SELECT er.ResourceID, er.ResourceType, COUNT(c.CampaignID) AS ImpactCount
2 FROM EnvironmentalResource er
3 JOIN Campaign c ON EXISTS (
4     SELECT 1
5     FROM Donation d
6     WHERE d.CampaignID = c.CampaignID
7     AND c.Description LIKE CONCAT('%', er.ResourceType, '%')
8 )
9 GROUP BY er.ResourceID, er.ResourceType
10 ORDER BY ImpactCount DESC
11 LIMIT 10;

```

Output:

Result Grid Filter Rows: Export: Wrap Cell Content:			
	ResourceID	ResourceType	ImpactCount
▶	6	Climate	1
	7	C Ocean	1
	4	Forest	1

10. Top 3 campaigns with the highest amount raised, along with the number of volunteers and donors who contributed to each campaign.

```
Query 1
1  SELECT c.CampaignID, c.Name, SUM(d.Amount) AS TotalAmountRaised,
2      COUNT(DISTINCT v.VolunteerID) AS TotalVolunteers,
3      COUNT(DISTINCT d.DonorID) AS TotalDonors
4  FROM Campaign c
5  JOIN Donation d ON c.CampaignID = d.CampaignID
6  JOIN Volunteer v ON d.DonorID = v.VolunteerID
7  GROUP BY c.CampaignID, c.Name
8  HAVING COUNT(v.VolunteerID) > 0 AND COUNT(d.DonorID) > 0
9  ORDER BY TotalVolunteers DESC
10 LIMIT 3;
```

Output:

	CampaignID	Name	TotalAmountRaised	TotalVolunteers	TotalDonors
▶	1	Climate Change Awareness Campaign	2770.00	13	13
	2	Ocean Conservation Campaign	2450.00	12	12
	3	Deforestation Prevention Campaign	2755.00	11	11

4. Database security and ethical issues

Trust, compliance, and corporate integrity depend on data security and ethical management. Security and ethical considerations should be at the top of the agenda when our business collects, stores, and processes sensitive data.

4.1 Ethical Considerations

1. **Data Privacy:** It is the most important thing in user right protection. It requires obedience to the privacy laws, such as CCPA and GDPR. Such rules will ensure that individuals have control over their information since all data collections, storage, and processing are transparently presented. Failure to observe this may incur serious legal consequences and damage to one's reputation (Barclays and Mahida, 2022).

2. **Data Minimization:** We should collect only the information that is needed for our goals. The more data there is, the higher the chance of security breach as well as a breach in ethics.
3. **Data Accuracy:** Data is to be refreshed and reviewed periodically to ensure relevance and integrity. Bad information lessens the value of the decisions supported by data and creates bad judgment.
4. **Transparency:** It creates an open line of communication with the data subjects regarding the collection, storage, and use of their data. It gives assurance to individuals about the processing of their information and helps in gaining trust.

4.2 Security Measures

The following security measures are recommended to protect data integrity, privacy, and availability:

1. **Permissions/Role-Based Access Control:** Limiting the user's role by using role-based access control hinders those users from accessing the data. This can be efficient while insider threats are reduced and less sensitive information exposure (*29th USENIX Conference on Security Symposium (SEC'20) : August 12 - 14, 2020, 2020*).
2. **Data Encryption:** Sensitive data should be encrypted when in transit and at rest. This prevents unauthorized access to data even if intercepted.
3. **MFA** - As the name suggests, it requires more than one method of verification. Hence, with MFA being much safer, there would be a reduced chance of any undesired access through compromised credentials.
4. **Regular Backups and Disaster Recovery:** Regular backups prevent data loss resulting from ransomware attacks, corruption, or even deletion. A disaster recovery plan should be put in place to ensure continuity and availability of data (Pien, 2020).
5. **Security of Network:** Firewalls, Intrusion Detection Systems, and Intrusion Prevention Systems are important to monitor and protect network access points.

4.3 Recommendations

Using Azure Security tools enhances our security and ensures compliance:

1. Azure Security Center: It provides vulnerability management, threat prevention, and compliance assessment. Security Center monitors the security posture of Azure resources while detecting potential vulnerabilities and providing actionable recommendations (Azevedo, 2021).
2. Azure Sentinel: Sentinel is a cloud-native SIEM application that does an analysis of security data from a lot of sources, helping in the identification, investigation, and response to security problems.
3. Azure Key Vault: Azure Key Vault is intended to securely store sensitive data, such as keys for encryption. It protects secrets and cryptographic keys, which are used to access data.

The steps would improve data security, reduce ethical issues, and ensure that the treatment of data is done in an ethical and transparent way by the company.

Conclusion

This paper ends with the description of the architecture of a robust database system for an environmental NGO that efficiently manages information related to volunteers, campaigns, donations, natural resources, and climatic impacts. The system is designed to provide effective tracking of key processes while data integrity can be ensured through normalization up to 3NF. The database allows for effective reporting and decision-making process through clear description of key entities and their relationships.

Security and ethical management are priorities, ensured through measures such as multi-factor authentication and encryption, role-based access control to sensitive data, and so on. It also guarantees trust and compliance through compliance with privacy laws, open data processing, and more (Ramaswamy and Sandhu, 2024).

The result of such a database system would be that it enables the mission and objectives of the NGO through increased capacity for managing activities, tracking performance, and helping their environmental programs, all while ensuring data security and ethical considerations.

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